fact that the emitter current I_E is due to holes and the base current I_B is due to electrons. However, the current through the external circuit is due to the flow of electrons.

EXAMPLE 10.5

In a transistor connected in the common base configuration, $\alpha = 0.95$, $I_E = 1\,mA$. Calculate the values of I_C and I_B.

Solution

$$\alpha = \frac{I_C}{I_E}$$

$$I_C = \alpha I_E = 0.95 \times 1 = 0.95\ mA$$

$$I_E = I_B + I_C$$

$$\therefore I_B = I_E - I_C = 1 - 0.95 = 0.05\ mA$$

10.4.3 Static Characteristics of Transistor in Common Emitter Mode

The know-how of certain parameters like the input resistance, output resistance, and current gain of a transistor are very important for the effective use of transistors in circuits. The circuit to study the static characteristics of an NPN transistor in the common emitter mode is given in Figure 10.30. The bias supply voltages V_{BB} and V_{CC} bias the base-emitter junction and collector-emitter junction respectively. The junction potential at the base-emitter is represented as V_{BE} and that at the collector-emitter as V_{CE}. The rheostats R_1 and R_2 are used to vary the base current and collector current respectively.

The static characteristics of the BJT are

i) Input characteristics
ii) Output characteristics
iii) Transfer characteristics

i) Input characteristics

Input characteristic curves give the relationship between the base current (I_B) and base to emitter voltage (V_{BE}) at constant collector to emitter voltage (V_{CE}) and are shown in Figure 10.31.

Initially, the collector to emitter voltage is set to a particular value (above 0.7 V to reverse bias the junction). Then the base-emitter voltage is increased in suitable steps and the corresponding base-current is recorded. A graph is plotted with V_{BE} along the x-axis and I_B along the y-axis. The procedure is repeated for different values of V_{CE}.

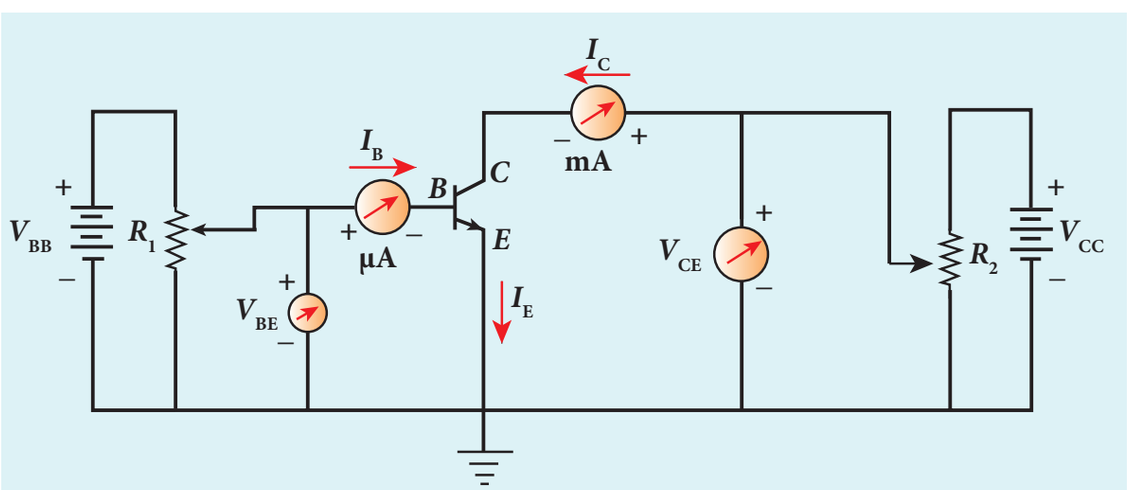


Figure 10.30 NPN transistor in common emitter configuration

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